# Wenyuan (Harry) Huang

whuang369@wisc.edu · whuang369.com · whuang369 · whuang369

## Education

**University of Wisconsin Madison** *Jan 2024 – May 2027 (Expected)*
*BS with Honor degree in Computer Science, Mathematics*

- GPA: 3.87/4.0
- **Coursework:** Advanced Topics in Reinforcement Learning, Nonlinear Optimization, Deep Learning for Computer Vision

## Experience

**Undergraduate Research Assistant** *Madison, WI*
*University of Wisconsin Madison - Computer Science Department* *Jan 2025 – Present*

- Conducting research in Reinforcement Learning (RL) with Prof. Josiah Hanna.
- Apply DRO (Distributionally Robust Optimization) to enhance the performance of algorithms such as PPO, DDPG, SA3, and RPO in multi-task RL, and extend its application to non-RL multi-task problems, including large language model (LLM) tasks.

**Machine Learning Peer Mentor** *Madison, WI*
*University of Wisconsin Madison - Computer Science Department* *Jan 2025 – Present*

- Holding office hours for 5 hours per week to help students with the course *Introduction to Artificial Intelligence*.
- Assisting students with implementing machine learning projects and assignments, including topics in Natural Language Processing, Deep Learning, and Reinforcement Learning.

## Projects

**Multi-task RL with Distributionally Robust Optimization** *MultiTaskRL*

- Applied Distributionally Robust Optimization to trajectory collections for various RL algorithms, including PPO, DDPG, and RPO.
- Implemented improved data collection and task sampling methods based on MTBench and CleanRL.
- Used MTRL techniques to reduce negative transfer between tasks, such as multi-head policy and one-hot task embedding.
- Planning to submit as a proceedings paper on ICML.

**Technology Blog in RL, Competitive Programming, Math** *My Blog*

- Writing blog posts covering topics in Reinforcement Learning, experiences in competitive programming, and mathematics.

## Honor & Awards

**Champion of ICPC North Central NA Regional Contest** Nov 2024
Ranked **1st among 96 ICPC teams** in the North Central North America region; the only team to solve 9 problems.

**Nicholas D. Yankaitis Memorial Scholarship** May 2025
Received one of the most prestigious awards recognizing academic excellence among undergraduate Computer Sciences majors.

## Technologies

**Languages:** Python, Java, C++, Markdown, LaTeX, HTML, CSS, JavaScript

**Technology skills:** Reinforcement Learning, Multi-task Learning, Deep Learning, Computer Vision, Algorithms, Data Structures